Number of Digits

Let $f(n)$ be the number of digits of n, where n is a positive integer.

Then $f(n)$ can be defined recursively as follows:

$$f(n) = 1 \text{ if } n < 10,$$

$$f(n) = 1 + f(\lfloor n/10 \rfloor) \text{ if } n \geq 10.$$

```
int numDigits( int number );

int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The number of digits of " << number << " is ";

   cout << numDigits( number ) << endl << endl;
}

int numDigits( int number )
{
   int count = 0;
   while( number > 0 )
   {
      count += 1;
      number /= 10;
   }
   return count;
}
```

```
int numDigits( int number )
{
   int count = 0;
   while( number > 0 )
   {
      count += 1;
      number /= 10;
   }
   return count;
}

int numDigits( int number )
{




}
```

```
int numDigits( int number )
{
   int count = 0;
   while( number > 0 )
   {
      count += 1;
      number /= 10;
   }
   return count;
}

int numDigits( int number )
{
   if( number < 10 )
      return 1;

   return                            ;
}
```

```
int numDigits( int number )
{
   int count = 0;
   while( number > 0 )
   {
      count += 1;
      number /= 10;
   }
   return count;
}

int numDigits( int number )
{
   if( number < 10 )
      return 1;

   return 1 + numDigits( number / 10 );
}
```

```
int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The number of digits of " << number << " is ";

   cout << numDigits( number ) << endl << endl;
}                               7
```

```
                                7
   int numDigits( int number )
   {
      if( number < 10 )
         return 1;

      return 1 + numDigits( number / 10 );
   }
```

```
int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The number of digits of " << number << " is ";

   cout << numDigits( number ) << endl << endl;
}                         78



                          78
   int numDigits( int number )
   {
      if( number < 10 )
         return 1;
                                     7
      return 1 + numDigits( number / 10 );
   }
```

```
int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The number of digits of " << number << " is ";

   cout << numDigits( number ) << endl << endl;
}                          789
```

```
                           789
   int numDigits( int number )
   {
      if( number < 10 )
         return 1;
                                      78
      return 1 + numDigits( number / 10 );
   }
```

```
int main()
{
    int number = 789;    789
    cout << numDigits( number ) << endl;
}
```

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

```
int main()
{
   int number = 789;      789
   cout << numDigits( number ) << endl;
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int main()
{
    int number = 789;
    cout << numDigits( number ) << endl;
}
```

789

789

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

78

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

789

```
int main()
{
   int number = 789;
   cout << numDigits( number ) << endl;
}
```

789

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

78

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int main()
{                              789
   int number = 789;
   cout << numDigits( number ) << endl;
}
```

```
                             789
int numDigits( int number )
{
   if( number < 10 )
      return 1;                        78
   return 1 + numDigits( number / 10 );
}
```

```
                                  78
int numDigits( int number )
{
   if( number < 10 )
      return 1;                            7
   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

789

```
int main()
{
   int number = 789;
   cout << numDigits( number ) << endl;
}
```

789

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

78

78

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

7

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int main()                789
{
    int number = 789;
    cout << numDigits( number ) << endl;
}

                          789
   int numDigits( int number )
   {
      if( number < 10 )
         return 1;                    78
      return 1 + numDigits( number / 10 );
   }

                                      78
           int numDigits( int number )
           {
              if( number < 10 )
                 return 1;                     7
              return 1 + numDigits( number / 10 );
           }

                                               7
                    int numDigits( int number )
                    {
                       if( number < 10 )
                          return 1; 1
                       return 1 + numDigits( number / 10 );
                    }
```

```
int main()
{
   int number = 789;
   cout << numDigits( number ) << endl;
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;
   return 1 + numDigits( number / 10 );
}
```

```
int main()
{                          789
   int number = 789;
   cout << numDigits( number ) << endl;
}

                           789
  int numDigits( int number )
  {
     if( number < 10 )
        return 1;                    78
     return 1 + numDigits( number / 10 );
  }         3       2
                                     78
         int numDigits( int number )
         {
            if( number < 10 )
               return 1;                        7
            return 1 + numDigits( number / 10 );
         }         2         1
                                                7
                  int numDigits( int number )
                  {
                     if( number < 10 )
                        return 1;  1
                     return 1 + numDigits( number / 10 );
                  }
```

```
int main()
{
    int number = 789;
    cout << numDigits( number ) << endl;
}
```

789

3

789

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

78

3 2

78

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

7

2 1

7

```
int numDigits( int number )
{
    if( number < 10 )
        return 1;
    return 1 + numDigits( number / 10 );
}
```

1

```
int numDigits( int number )
{
   if( number < 10 )
      return 1;

   return 1 + numDigits( number / 10 );
}
```

```
int numDigits( int number )
{
   if( number > 9 )
      return 1 + numDigits( number / 10 );

   return 1;
}
```

```
int main()
{
    int n = 789;
    cout << f( n ) << endl;
}
```

$\boxed{789}$

$\boxed{789}$

```
int f( int n )
{
    if( n < 10 ) return 1;
    return 1 + f( n / 10 );
}
```

78

$\boxed{789}$

```
int f( int n )
{
    if( n > 9 ) return 1 + f( n / 10 );
    return 1;
}
```

78

$\boxed{78}$

```
int f( int n )
{
    if( n < 10 ) return 1;
    return 1 + f( n / 10 );
}
```

7

$\boxed{78}$

```
int f( int n )
{
    if( n > 9 ) return 1 + f( n / 10 );
    return 1;
}
```

7

$\boxed{7}$

$\boxed{7}$

Summing Digits

Let $f(n)$ be the sum of digits of n, where n is a positive integer.

Then $f(n)$ can be defined recursively as follows:

$$f(n) = n \text{ if } n < 10,$$

$$f(n) = n \bmod 10 + f(\lfloor n/10 \rfloor) \text{ if } n \geq 10.$$

```
int sumDigits( int number );

int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The sum of digits of " << number << " is ";

   cout << sumDigits( number ) << endl << endl;
}

int sumDigits( int number )
{
   int sum = 0;
   while( number > 0 )
   {
      sum += number % 10;
      number /= 10;
   }
   return sum;
}
```

```cpp
int sumDigits( int number );

int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The sum of digits of " << number << " is ";

   cout << sumDigits( number ) << endl << endl;
}

int sumDigits( int number )
{




}
```

```
int sumDigits( int number );

int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The sum of digits of " << number << " is ";

   cout << sumDigits( number ) << endl << endl;
}

int sumDigits( int number )
{
   if( number < 10 )
      return number;

   return sumDigits(             ) +            ;
}
```

```cpp
int sumDigits( int number );

int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The sum of digits of " << number << " is ";

   cout << sumDigits( number ) << endl << endl;
}

int sumDigits( int number )
{
   if( number < 10 )
      return number;

   return sumDigits( number / 10 ) + number % 10;
}
```

```
int main()                789
{
   int number = 789;
   cout << sumDigits( number ) << endl;
}
             24
```

```
                      789
  int sumDigits( int number )
  {
     if( number < 10 )
        return number;         78        24        9
     return sumDigits( number / 10 ) + number % 10;
  }
              15
```

```
                                   78
         int sumDigits( int number )
         {
            if( number < 10 )
               return number;         7        15        8
            return sumDigits( number / 10 ) + number % 10;
         }
                     7
```

```
                                        7
               int sumDigits( int number )
               {
                  if( number < 10 )
                     return number;  7
                  return sumDigits( number / 10 ) + number % 10;
               }
```

```
int sumDigits( int number )
{
   if( number < 10 )
      return number;

   return sumDigits( number / 10 ) + number % 10;
}
```

```
int sumDigits( int number )
{
   if( number > 9 )
      return sumDigits( number / 10 ) + number % 10;

   return number;
}
```

```
int main()
{
   int n = 789;
   cout << f( n ) << endl;
}
             24
```

```
789
int f( int n )
{
   if( n < 10 ) return n;
   return f( n / 10 ) + n % 10;
}
             78  15   9
```

```
78
int f( int n )
{
   if( n < 10 ) return n;
   return f( n / 10 ) + n % 10;
}
             7        8
```

```
7
```

```
789
int f( int n )
{
   if( n > 9 ) return f( n / 10 ) + n % 10;
   return n;
}
                         78  15   9
```

```
78
int f( int n )
{
   if( n > 9 ) return f( n / 10 ) + n % 10;
   return n;
}
                         7        8
```

```
7
```

Prints Digits in the Reverse Order

```
void reverse( int number );

int main()
{
   int number;
   cin >> number;
   reverse( number );
}

void reverse( int number )
{
   while( number > 0 )
   {
      cout << number % 10;
      number /= 10;
   }
}
```

```
void reverse( int number );

int main()
{
   int number;
   cin >> number;
   reverse( number );
}

void reverse( int number )
{



}
```

```
void reverse( int number );

int main()
{
   int number;
   cin >> number;
   reverse( number );
}

void reverse( int number )
{
   cout <<            ;
   if( number > 9 )
      reverse(             );
}
```

```cpp
void reverse( int number );

int main()
{
   int number;
   cin >> number;
   reverse( number );
}

void reverse( int number )
{
   cout << number % 10;
   if( number > 9 )
      reverse( number / 10 );
}
```

```
int main()
{
   int number = 789;
   reverse( number );
}
```

Output

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
int main()
{
   int number = 789;
   reverse( number );
}
```

789

Output

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
int main()
{
   int number = 789;
   reverse( number );
}
```

789

789

9

Output

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
int main()
{
   int number = 789;
   reverse( number );
}
```

789

789

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

78

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

9

Output

```
int main()
{
   int number = 789;
   reverse( number );
}
```

789

789

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

78

78

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
98
```

Output

```
int main()
{
   int number = 789;
   reverse( number );
}
```

789

98

Output

789

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

78

78

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

7

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

```
int main()
{
   int number = 789;
   reverse( number );
}
```

789

789

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

78

78

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

7

7

```
void reverse( int number )
{
   cout << number % 10;
   if( number > 9 ) reverse( number / 10 );
}
```

987

Output

Prints Digits of An Integer

```cpp
int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The digits of " << number << ": ";

   digits( number );

   cout << endl << endl;
}

void digits( int number )
{




}
```

```
int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The digits of " << number << ": ";

   digits( number );

   cout << endl << endl;
}

void digits( int number )
{
   if( number > 9 )
      digits(             );

   cout <<            ;
}
```

```
int main()
{
   srand( static_cast< int >( time( 0 ) ) );

   int number = rand();

   cout << "The digits of " << number << ": ";

   digits( number );

   cout << endl << endl;
}

void digits( int number )
{
   if( number > 9 )
      digits( number / 10 );

   cout << number % 10;
}
```

```
int main()
{
    int number = 789;
    digits( number );
}
```

789

Output

```
void digits( int number )
{
    if( number > 9 ) digits( number/10 );
    cout << number % 10;
}
```

```
void digits( int number )
{
    if( number > 9 ) digits( number/10 );
    cout << number % 10;
}
```

```
void digits( int number )
{
    if( number > 9 ) digits( number/10 );
    cout << number % 10;
}
```

```
int main()
{
    int number = 789;
    digits( number );
}
```

789

Output

789

```
void digits( int number )
{
    if( number > 9 ) digits( number/10 );
    cout << number % 10;
}
```

78

```
void digits( int number )
{
    if( number > 9 ) digits( number/10 );
    cout << number % 10;
}
```

```
void digits( int number )
{
    if( number > 9 ) digits( number/10 );
    cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
```

789

Output

789

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

78

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
```

789

Output

789

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

78

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
            789
```

Output

```
            789
void digits( int number )
{
   if( number > 9 ) digits( number / 10 );
   cout << number % 10;
}
```

```
                        78
void digits( int number )
{
   if( number > 9 ) digits( number / 10 );
   cout << number % 10;                   7
}
```

```
void digits( int number )
{
   if( number > 9 ) digits( number / 10 );
   cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
```

789

Output

789

```
void digits( int number )
{
   if( number > 9 ) digits( number / 10 );
   cout << number % 10;
}
```

78

```
void digits( int number )
{
   if( number > 9 ) digits( number / 10 );
   cout << number % 10;
}
```

7

```
void digits( int number )
{
   if( number > 9 ) digits( number / 10 );
   cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
```

789

Output

789

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

78

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

7

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
```

789

Output

7

789

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

78

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

7

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

```
int main()
{
    int number = 789;
    digits( number );
}
```

789

78

Output

789

```
void digits( int number )
{
    if( number > 9 ) digits( number / 10 );
    cout << number % 10;
}
```

78

```
void digits( int number )
{
    if( number > 9 ) digits( number / 10 );
    cout << number % 10;
}
```

7

```
void digits( int number )
{
    if( number > 9 ) digits( number / 10 );
    cout << number % 10;
}
```

```
int main()
{
   int number = 789;
   digits( number );
}
```

789

Output

789

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

78

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

7

```
void digits( int number )
{
   if( number > 9 ) digits( number/10 );
   cout << number % 10;
}
```

Prints Digits of An Integer Forward and Backward

```
void digits( int number )
{
   if( number > 9 )
      digits( number / 10 );

   cout << number % 10;
}

void reverse( int number )
{
   cout << number % 10;

   if( number > 9 )
      reverse( number / 10 );
}
```

```
void digits( int n )
{
   if( n > 9 ) digits( n / 10 );
   cout << n % 10;
}

void reverse( int n )
{
   cout << n % 10;
   if( n > 9 ) reverse( n / 10 );
}
```

```
int main()
{
    int number = 789;
    reverse( number );
}
                789
```

```
                789
void reverse( int n )
{             9
    cout << n % 10;              78
    if( n > 9 ) reverse( n / 10 );
}
```

```
                78
void reverse( int n )
{             8
    cout << n % 10;              7
    if( n > 9 ) reverse( n / 10 );
}
```

```
                7
void reverse( int n )
{             7
    cout << n % 10;
    if( n > 9 ) reverse( n / 10 );
}
```

```
int main()
{
    int number = 789;
    digits( number );
}
                789
```

```
                789
void digits( int n )
{
    if( n > 9 ) digits( n / 10 );
    cout << n % 10;              78
}         9
```

```
                78
void digits( int n )
{
    if( n > 9 ) digits( n / 10 );
    cout << n % 10;              7
}         8
```

```
                7
void digits( int n )
{
    if( n > 9 ) digits( n / 10 );
    cout << n % 10;
}         7
```